Manashvi Kainturia
500122403
Data Science B4 __/__/____

Assignment 1

CL & NLP

Ques 1 Given $x_t = 0.5$, $h_{t-1} = 0.2$, $C_{t-1} = 0.4$

Step 1 Forget Gate f_t

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$
$$= \sigma((0.7)(0.5) + (0.3)(0.2) + 0.1)$$
$$= \sigma(0.35 + 0.06 + 0.1)$$
$$= \sigma(0.5100)$$

$$f_t = \frac{1}{1 + e^{-0.5100}}$$

$$f_t = 0.6248$$

Step 2 Input Gate i_t

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$
$$= \sigma((0.6)(0.5) + (0.4)(0.2) + 0)$$
$$= \sigma(0.30 + 0.08)$$
$$= \sigma(0.3800)$$

$$= \frac{1}{1 + e^{-0.3800}}$$

$$i_t = 0.5939$$

__/__/____

Step 3 candidate cell state $\tilde{C}_t$

$$\tilde{C}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$
$$= \tanh((0.9)(0.5) + (0.2)(0.2) + 0)$$
$$= \tanh(0.4900)$$
$$= 0.4542$$

Step 4 New cell state C_t

$$C_t = f_t C_{t-1} + i_t \tilde{C}_t$$
$$= (0.6248)(0.4) + (0.5939)(0.4542)$$
$$= 0.2499 + 0.2697$$
$$= 0.5196$$

Step 5 Output gate O_t

$$O_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$
$$= \sigma((0.5)(0.5) + (0.5)(0.2) + 0.1)$$
$$= \sigma(0.4500)$$
$$= \frac{1}{1 + e^{-0.4500}}$$

$$O_t = 0.6106$$

__/__/____

Step 6 Hidden state h_t

$$h_t = o_t \tanh(C_t)$$
$$= 0.6106 \tanh(0.5196)$$
$$= 0.6106 \times 0.4773$$
$$h_t \approx 0.2995$$

Ques 2 Weight for input = $32 \times 10 = 320$
Weight for hidden = $32 \times 32 = 1024$

(a) Single bias convention

$$320 + 1024 + 32 = 1376$$
$$4(1376) = 5504$$

(b) Two bias convention

$$320 + 1024 + 32 + 32 = 1408$$
$$4(1408) = 5632$$

Ques 3 (a) The gradient scaling is : f^k

After 10 steps → $(0.9)^{10}$

$= 0.3487$

After 50 steps $= (0.9)^{50}$

$= 0.0052$

(b) $k = 50$

$f = 0.5$

$(0.5)^{50} = 8.8816 \times 10^{-16}$

for $f = 0.99$

$(0.99)^{50} = 0.6050$

(c) For a dependency 50 times steps back, o forget gate for 0.9 reduces the gradient to only about 0.527., making learning long term dependencies difficult.

LSTM, help because their cell state provide a relatively direct path for information & gradient & the forget gate can learn values close to 1, greatly reducing vanishing compared with a vanilla RNN

__/__/____

Ques 4 (a) No. of hourly records

1 year = 365 × 24

= 8760

3 years = 3 × 8760

= 26280

(b) No. of input / output windows

each window requires = 168 + 24

= 192 h

with stride 1 : Nwindows = N - L - H + 1

= 26280 - 168 - 24 + 1

= 26089

(c) Tensor Shapes

each input window contains - 168 times steps

3 features

For batch size 64 x : (64, 168, 3)

The model predicts the next 24 h

Therefore, y = (64, 24)